EMPLOYEE ATTRITION SQL ANALYSIS REPORT

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1. Project Overview

In this project, I analyzed an HR dataset to uncover the key factors influencing employee attrition. My goal was to transform raw employee data into actionable insights that could help HR departments reduce turnover, improve retention, and enhance workforce planning.

All data cleaning, transformation, and analysis were done using **SQL**, focusing on efficiency and real-world HR metrics such as attrition rate, average income, and tenure.

2. Dataset Description

The dataset contains employee information across various dimensions including:

- Employee demographics (Age, Gender, Marital Status)
- Job-related data (Department, Job Role, Job Satisfaction)
- Compensation metrics (Monthly Income)
- Tenure information (Years at Company, Years with current Manager, Total working years)
- Attrition status (Yes/No)

3. Objectives

1. Identify overall and departmental attrition rates.
2. Compare average monthly income of employees who left.
3. Analyze the relationship between job satisfaction and attrition.
4. Explore tenure patterns to see how years at the company affect attrition.
5. Provide data-driven recommendations to help HR make informed retention strategies.

4. SQL Analysis

Below are the key SQL queries I wrote and their respective insights.

A. Total Employee Count

```
SELECT COUNT(*) AS Total_Employees
FROM attrition
;
```

Insight: The total number of employees forms the baseline for calculating attrition and other averages.

B. Total Attrition Count

```
SELECT COUNT(*) AS Employees__who_left
FROM attrition2
WHERE Attrition = 'Yes'
;
```

Insight: This gives the total number of employees who have left the company. Useful for tracking turnover.

C. Attrition Rate

```
SELECT
    (SUM(CASE WHEN Attrition='Yes' THEN 1 ELSE 0 END) * 100.0 / COUNT(*) ) AS Attrition_Rate
FROM attrition2
;
```

Insight: This provides the company's overall attrition rate, a key HR KPI that signals employee satisfaction and engagement.

D. Average Monthly Income (by Attrition Status)

```
SELECT
    ROUND(AVG(Monthly_Income),2) AS Avg_Monthly_Income
FROM attrition2
WHERE Attrition = 'Yes';
```

Insight: Employees who left tend to have lower average monthly income compared to those who stayed, income appears to influence attrition.

E. Average Tenure (Years at Company)

```
SELECT
ROUND(AVG(Years_At_Company),2) AS Avg_tenure_at_company
FROM Attrition2
WHERE Attrition = 'Yes';
```

Insight: Employees who left typically had shorter tenures, suggesting early-stage disengagement.

F. Department-wise Attrition

```
SELECT Department,
        COUNT(*) AS Total,
        SUM(CASE WHEN Attrition='Yes' THEN 1 ELSE 0 END) Employees_Left,
    ROUND(SUM(CASE WHEN Attrition='Yes' THEN 1 ELSE 0 END) * 100.0 / COUNT(*),2) Attrition_Rate
FROM attrition2
GROUP BY Department
ORDER BY Attrition_Rate DESC;
```

Insight: The **Sales** department showed the highest attrition, followed by **Human Resources**, while **Research & Development** retained employees more effectively.

G. Job Satisfaction vs Attrition

```
 SELECT
    Job_Satisfaction,
    COUNT(CASE WHEN Attrition = 'Yes' THEN 1 END) AS Employees_Left,
    COUNT(*) AS Total_Employees,
    ROUND((COUNT(CASE WHEN Attrition = 'Yes' THEN 1 END) * 100.0 / COUNT(*)), 2) AS Attrition_Rate
FROM attrition2
GROUP BY Job_Satisfaction
ORDER BY Job_Satisfaction;
```

Insight: Lower job satisfaction scores correlate strongly with higher attrition rates, a key actionable insight for HR engagement programs.

H. Gender-Based Attrition

```
SELECT Gender,
ROUND(SUM(CASE WHEN Attrition='Yes' THEN 1 ELSE 0 END) * 100.0 / COUNT(*),2) Attrition_Rate
FROM attrition2
GROUP BY Gender
ORDER BY Attrition_Rate DESC;
 ;
```

Insight: Male employees show slightly higher attrition compared to females, suggesting possible work-life or role satisfaction gaps.

I. Age Band vs Attrition

```sql
SELECT
    CASE
    WHEN Age BETWEEN 18 AND 25 THEN
    '18-25'
    WHEN Age BETWEEN 26 AND 35 THEN
    '26-35'
    WHEN Age BETWEEN 36 AND 45 THEN
    '36-45'
    WHEN Age BETWEEN 46 AND 55 THEN
    '46-55'
    ELSE '56+'
    END AS Age_Group,
    ROUND(SUM(CASE WHEN Attrition='Yes' THEN 1 ELSE 0 END) * 100.0 / COUNT(*),2) Attrition_Rate
FROM attrition2
GROUP BY  Age_Group
ORDER BY Attrition_Rate DESC
 ;
```

Insight: Younger employees (especially 26–35) tend to leave more frequently, often for better opportunities or faster growth.

5. Summary of Key Insights

Metric	Insight
Attrition Rate	Around 16–20% overall — moderate but concerning.
Top Risk Department	Sales had the highest attrition rate.
Income Influence	Employees earning less tend to leave more often.
Job Satisfaction	Strong negative correlation with attrition.
Tenure Pattern	Most exits occur within the first 3–5 years.
Demographic Factor	Younger employees (26–35) and males slightly more prone to leave.

6. Recommendations

1. **Improve Retention in Sales:** Conduct stay interviews and increase growth incentives.
2. **Compensation Review:** Evaluate pay structure to close income gaps.
3. **Engagement Programs:** Target employees with low satisfaction scores for career development support.
4. **Early Tenure Support:** Strengthen onboarding and mentorship to reduce early exits.

7. Tools Used

- **SQL (Structured Query Language)** for data cleaning, analysis, and querying.
- **Excel** (for secondary dashboard visualization).
- **PDF/Documentation Tools** for reporting.

8. Conclusion

This SQL analysis provided a clear, data-backed understanding of employee attrition patterns. By integrating structured queries with HR domain logic, I was able to identify where, why, and among whom attrition is highest, helping decision-makers take precise retention actions.